20. Which of the following lanthanoid ions is diamagnetic?

a) Eu^{2+} b) Yb^{2+}

c) Ce^{2+} d) Sm^{2+}

21. Which of the following oxidation states is most common among the lanthanoids?

a) 4 b) 2

c) 5 d) 3

22. Assertion : Ce^{4+} is used as an oxidizing agent in volumetric analysis.
Reason: Ce^{4+} has the tendency of attaining +3 oxidation state.
a) Both assertion and reason are true and reason is the correct explanation of assertion.
b) Both assertion and reason are true but reason is not the correct explanation of assertion.
c) Assertion is true but reason is false.
d) Both assertion and reason are false.

23. The most common oxidation state of actinoids is

a) +2 b) +3

c) +4 d) +6

24. The actinoid elements which show the highest oxidation state of +7 are

a) Np, Pu ,Am b) U, Fm, Th

c) U, Th, Md d) Es, No, Lr

25. Which one of the following is not correct?

a) $La(OH)_2$ is less basic than $Lu(OH)_3$

b) In lanthanoid series ionic radius of Ln^{3+} ions decreases

c) La is actually an element of transition metal series rather than lanthanide series

d) Atomic radii of Zr and Hf are same because of lanthanide contraction

Answer the following questions:

1. What are transition metals? Give four examples.
2. Explain the oxidation states of 3d series elements.
3. What are inner transition elements?
4. Justify the position of lanthanides and actinides in the periodic table.
5. What are actinides? Give three examples.
6. Why Gd^{3+} is colourless?
7. Explain why compounds of Cu^{2+} are coloured but those of Zn^{2+} are colourless.
8. Describe the preparation of potassium dichromate.

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